

torch.sum#

<https://pytorch.org/docs/stable/generated/torch.sum.html#torch.sum>

Description:

Returns the sum of all elements in the input tensor. `input` (Tensor) – the input tensor. `dtype` (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None. Use the dtype argument if you need the result in a specific tensor type. Otherwise, the result type may be automatically promoted (e.g., from torch.int32 to torch.int64).

Function Signature

```
torch.sum(input, *, dtype=None) → Tensor#
```

Parameters

Keyword Arguments

```
torch.sum(input, dim, keepdim=False, *, dtype=None) → Tensor
```

Parameters

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Keyword

`dtype` (`torch.dtype`, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to `dtype` before the operation is performed. This is useful for preventing data type overflows. Default: None.

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Code Examples

```
>>> a = torch.randn(1, 3)
>>> a
tensor([[ 0.1133, -0.9567,  0.2958]])
>>> torch.sum(a)
tensor(-0.5475)
```

```
>>> a = torch.randn(4, 4)
>>> a
tensor([[ 0.0569, -0.2475,  0.0737, -0.3429],
        [-0.2993,  0.9138,  0.9337, -1.6864],
        [ 0.1132,  0.7892, -0.1003,  0.5688],
        [ 0.3637, -0.9906, -0.4752, -1.5197]])
>>> torch.sum(a, 1)
tensor([-0.4598, -0.1381,  1.3708, -2.6217])
>>> b = torch.arange(4 * 5 * 6).view(4, 5, 6)
>>> torch.sum(b, (2, 1))
tensor([ 435., 1335., 2235., 3135.])
```

Notes & Warnings

NOTE Use the `dtype` argument if you need the result in a specific tensor type. Otherwise, the result type may be automatically promoted (e.g., from `torch.int32` to `torch.int64`).

Detailed Documentation

Documentation

Returns the sum of all elements in the input tensor.

`input (Tensor)` – the input tensor.

`dtype (torch.dtype, optional)` – the desired data type of returned tensor. If specified, the input tensor is casted to `dtype` before the operation is performed. This is useful for preventing data type overflows. Default: `None`.

Use the `dtype` argument if you need the result in a specific tensor type. Otherwise, the result type may be automatically promoted (e.g., from `torch.int32` to `torch.int64`).

Returns the sum of each row of the input tensor in the given dimension `dim`. If `dim` is a list of dimensions, reduce over all of them.

If `keepdim` is `True`, the output tensor is of the same size as input except in the dimension(s) `dim` where it is of size 1. Otherwise, `dim` is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 (or `len(dim)`) fewer dimension(s).

`input (Tensor)` – the input tensor.

`dim (int or tuple of ints, optional)` – the dimension or dimensions to reduce. If `None`, all dimensions are reduced.

`keepdim (bool, optional)` – whether the output tensor has `dim` retained or not. Default: `False`.

`dtype (torch.dtype, optional)` – the desired data type of returned tensor. If specified, the

input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None.